# Evaluating multi-season occupancy models with autocorrelation fitted to heterogeneous datasets

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# Supporting Information F - Results of multi-season site occupancy model for the remaining four butterfly species

This file shows the maps and tables (coefficients, RHat and Prior-Posterior Overlap) resulting from the occupancy analyses for four butterfly species (not shown in the main text): the false ringlet *Coenonympha oedippus* (Fabricius, 1787), the marsh fritillary *Euphydryas aurinia* (Rottemburg, 1775), the small copper *Lycaena phlaeas* (Linnaeus, 1761), and the meadow brown *Maniola jurtina* (Linnaeus, 1758). We did analyses at the Bordeaux + 10 km buffer scale, and used weakly informative priors for the spatial decay parameter $\phi$ in the occupancy models (as described in the main text).

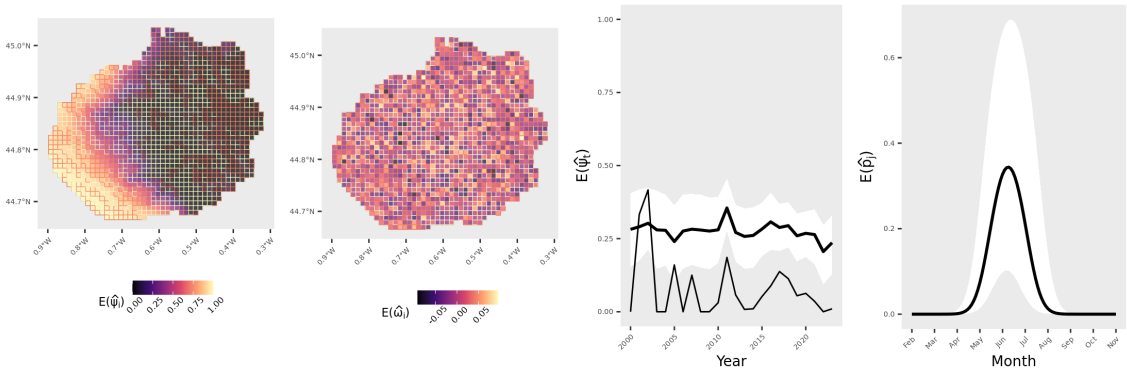


Fig. F.1: Distribution of the false ringlet *Coenonympha oedippus* in Southwest France (buffer of 10 km$^2$ around Bordeaux). Results for the model with NGG=15 and weakly informative prior for $\phi$. Values plotted in the occupancy map (left) were calculated as $\mathbb{E}(\hat{\psi}_i) = \frac{1}{T}\sum_{t=1}^{T} \hat{\psi}_{it}$, and refer to in-sample estimates for sites with data (cells with yellow borders) and out-of-sample predictions for missing cells (those with red borders). The spatial random effect $\mathbb{E}(\hat{\omega}_i)$ was also derived from in-sample estimates and out-of-sample predictions from the model. Trend plots show the proportion of occupied sites over the 24 years of data, calculated as $\mathbb{E}(\hat{\psi}_t) = \frac{1}{I}\sum_{i=1}^{I} \hat{\psi}_{it}$. The naive occupancy trend (number of sites with detection divided by the total number of sampled sites per year) is depicted by the thinner line. Detection probability over months (secondary periods) was also derived from in-sample predictions. The black line depicts the average detection trend, and the white bands depict the upper and lower bounds of the 95% credible interval. No $1 \times 1$ km cell was 100% covered by water (the maximum was 93%). Parameter values derived from 1,500 posterior distribution draws.

Table F.1: Estimated parameters of the multi-season site-occupancy model with spatial and temporal random effects fitted to data of the false ringlet *Coenonympha oedippus* (Fabricius, 1787). Results for models with 15 spatial neighbors, and weak or informative $\phi$ priors. PPO: Prior-Posterior Overlap. PPO: Prior-Posterior Overlap. Abbreviation of covariates = LAT: Latitude, LON: Longitude, ELEV= Elevation, LAND= non-water land cover, GRASS: Cover of natural grasslands, OBS: Number of observers, MON: survey month. The mean, 95% Credible Intervals, and Rhat were calculated using the 1,500 posterior distribution draws of each parameter.

| Parameter | Mean | LCI | UCI | RHat | Overlap |
|---|---|---|---|---|---|
| $\beta_0$ | -4.623 | -6.220 | -3.069 | 1.058 | 6.7 |
| $\beta_{LAT}$ | 0.227 | -0.692 | 1.124 | 1.086 | 45.2 |
| $\beta_{LAT^2}$ | 1.571 | 0.877 | 2.357 | 1.189 | 27.3 |
| $\beta_{LON}$ | -3.781 | -4.933 | -2.716 | 1.088 | 8.7 |
| $\beta_{ELEV}$ | 2.108 | 1.043 | 3.278 | 1.016 | 29.1 |
| $\beta_{NON-WATER}$ | -0.941 | -3.757 | 1.861 | 1.146 | 75.0 |
| $\beta_{NON-WATER^2}$ | -1.555 | -4.551 | 0.756 | 2.621 | 60.8 |
| $\beta_{URBAN}$ | -0.693 | -2.792 | 0.552 | 1.173 | 62.7 |
| $\sigma^2$ | 1.154 | 0.354 | 2.767 | 1.570 | 83.2 |
| $\phi$ | 32.184 | 4.555 | 59.016 | 1.003 | 8.0 |
| $\sigma_T^2$ | 0.986 | 0.233 | 3.108 | 1.013 | 86.6 |
| $\rho$ | 0.072 | -0.311 | 0.420 | 1.022 | 43.7 |
| $\alpha_0$ | -0.721 | -2.324 | 0.780 | 1.161 | 61.0 |
| $\alpha_{NON-WATER}$ | 0.403 | -2.798 | 3.241 | 1.399 | 87.6 |
| $\alpha_{LAT}$ | 0.382 | 0.039 | 0.707 | 1.032 | 19.1 |
| $\alpha_{OBS}$ | 0.064 | -0.048 | 0.189 | 1.007 | 8.5 |
| $\alpha_{MON}$ | -1.733 | -2.732 | -0.743 | 1.005 | 33.2 |
| $\alpha_{MON^2}$ | -9.937 | -12.261 | -7.929 | 1.019 | 0.0 |

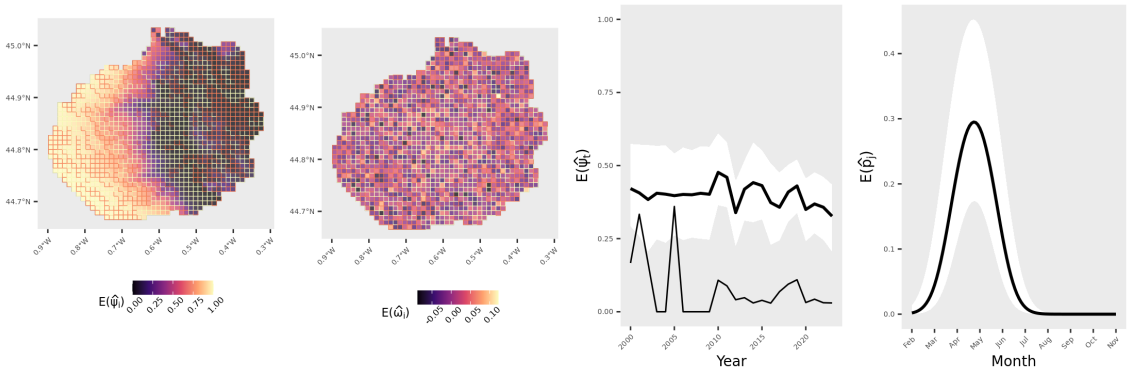


Fig. F.2: Distribution of the marsh fritillary *Euphydryas aurinia* in Southwest France (buffer of 10 km$^2$ around Bordeaux). Results for the model with NGG=15 and weakly informative prior for $\phi$. Values plotted in the occupancy map (left) were calculated as $\mathbb{E}(\hat{\psi}_i) = \frac{1}{T}\sum_{t=1}^{T} \hat{\psi}_{it}$, and refer to in-sample estimates for sites with data (cells with yellow borders) and out-of-sample predictions for missing cells (those with red borders). The spatial random effect $\mathbb{E}(\hat{\omega}_i)$ was also derived from in-sample estimates and out-of-sample predictions from the model. Trend plots show the proportion of occupied sites over the 24 years of data, calculated as $\mathbb{E}(\hat{\psi}_t) = \frac{1}{I}\sum_{i=1}^{I} \hat{\psi}_{it}$. The naive occupancy trend (number of sites with detection divided by the total number of sampled sites per year) is depicted by the thinner line. Detection probability over months (secondary periods) was also derived from in-sample predictions. The black line depicts the average detection trend, and the white bands depict the upper and lower bounds of the 95% credible interval. No $1 \times 1$ km cell was 100% covered by water (the maximum was 93%). Parameter values derived from 1,500 posterior distribution draws.

Table F.2: Estimated parameters of the multi-season site-occupancy model with spatial and temporal random effects fitted to data of the marsh fritillary *Euphydryas aurinia* (Rottemburg, 1775). Results for models with 15 spatial neighbors, and weak or informative $\phi$ priors. PPO: Prior-Posterior Overlap. PPO: Prior-Posterior Overlap. Abbreviation of covariates = LAT: Latitude, LON: Longitude, ELEV= Elevation, LAND= non-water land cover, GRASS: Cover of natural grasslands, OBS: Number of observers, MON: survey month. The mean, 95% Credible Intervals, and Rhat were calculated using the 1,500 posterior distribution draws of each parameter.

| Parameter | Mean | LCI | UCI | RHat | Overlap |
|---|---|---|---|---|---|
| $\beta_0$ | -2.122 | -3.512 | -0.821 | 1.011 | 30.8 |
| $\beta_{LAT}$ | 0.337 | -0.494 | 1.227 | 1.019 | 44.8 |
| $\beta_{LAT^2}$ | 1.194 | 0.585 | 1.938 | 1.217 | 28.5 |
| $\beta_{LON}$ | -3.637 | -4.952 | -2.417 | 1.257 | 9.3 |
| $\beta_{ELEV}$ | 1.767 | 0.770 | 2.916 | 1.339 | 31.5 |
| $\beta_{NON-WATER}$ | -0.675 | -2.706 | 1.285 | 1.008 | 72.9 |
| $\beta_{NON-WATER^2}$ | -0.677 | -1.681 | 0.203 | 1.022 | 44.8 |
| $\beta_{URBAN}$ | -1.135 | -4.232 | 0.227 | 2.635 | 60.2 |
| $\sigma^2$ | 1.297 | 0.278 | 3.760 | 2.671 | 90.7 |
| $\phi$ | 31.397 | 4.437 | 58.805 | 1.015 | 8.5 |
| $\sigma_T^2$ | 1.360 | 0.262 | 4.194 | 1.037 | 69.3 |
| $\rho$ | 0.061 | -0.317 | 0.422 | 1.042 | 44.8 |
| $\alpha_0$ | -3.091 | -4.271 | -1.964 | 1.002 | 15.3 |
| $\alpha_{NON-WATER}$ | -1.100 | -2.665 | 0.174 | 1.011 | 52.8 |
| $\alpha_{LAT}$ | 0.014 | -0.270 | 0.299 | 1.014 | 17.6 |
| $\alpha_{OBS}$ | 0.057 | -0.035 | 0.155 | 1.020 | 6.8 |
| $\alpha_{MON}$ | -7.264 | -8.947 | -5.699 | 1.060 | 0.4 |
| $\alpha_{MON^2}$ | -5.943 | -7.383 | -4.581 | 1.036 | 1.3 |

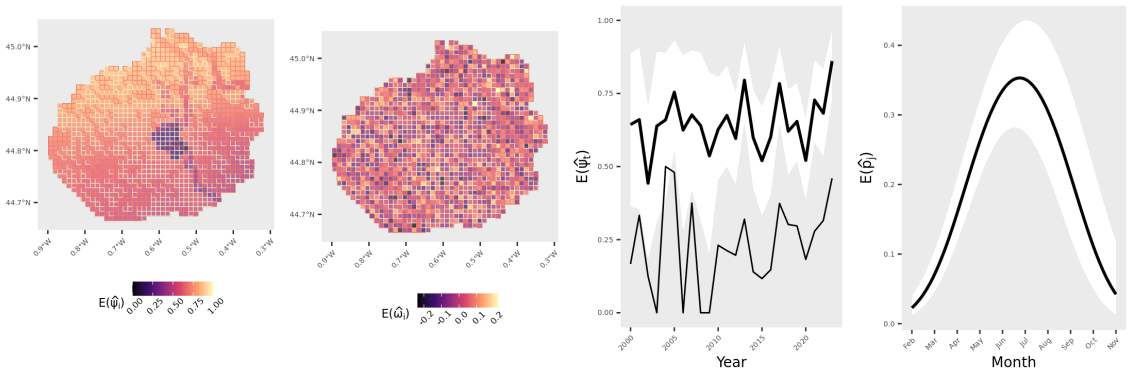


Fig. F.3: Distribution of the small copper *Lycaena phlaeas* in Southwest France (buffer of 10 km$^2$ around Bordeaux). Results for the model with NGG=15 and weakly informative prior for $\phi$. Values plotted in the occupancy map (left) were calculated as $\mathbb{E}(\hat{\psi}_i) = \frac{1}{T}\sum_{t=1}^{T} \hat{\psi}_{it}$, and refer to in-sample estimates for sites with data (cells with yellow borders) and out-of-sample predictions for missing cells (those with red borders). The spatial random effect $\mathbb{E}(\hat{\omega}_i)$ was also derived from in-sample estimates and out-of-sample predictions from the model. Trend plots show the proportion of occupied sites over the 24 years of data, calculated as $\mathbb{E}(\hat{\psi}_t) = \frac{1}{I}\sum_{i=1}^{I} \hat{\psi}_{it}$. The naive occupancy trend (number of sites with detection divided by the total number of sampled sites per year) is depicted by the thinner line. Detection probability over months (secondary periods) was also derived from in-sample predictions. The black line depicts the average detection trend, and the white bands depict the upper and lower bounds of the 95% credible interval. No $1 \times 1$ km cell was 100% covered by water (the maximum was 93%). Parameter values derived from 1,500 posterior distribution draws.

Table F.3: Estimated parameters of the multi-season site-occupancy model with spatial and temporal random effects fitted to data of the small copper *Lycaena phlaeas* (Linnaeus, 1761). Results for models with 15 spatial neighbors, and weak or informative $\phi$ priors. PPO: Prior-Posterior Overlap. PPO: Prior-Posterior Overlap. Abbreviation of covariates = LAT: Latitude, LON: Longitude, ELEV= Elevation, LAND= non-water land cover, GRASS: Cover of natural grasslands, OBS: Number of observers, MON: survey month. The mean, 95% Credible Intervals, and Rhat were calculated using the 1,500 posterior distribution draws of each parameter.

| Parameter | Mean | LCI | UCI | RHat | Overlap |
|---|---|---|---|---|---|
| $\beta_0$ | 1.142 | -0.045 | 2.338 | 1.068 | 45.9 |
| $\beta_{LAT}$ | 1.052 | 0.103 | 1.971 | 1.190 | 40.3 |
| $\beta_{LAT^2}$ | 0.258 | -0.475 | 0.970 | 1.106 | 36.6 |
| $\beta_{LON}$ | -0.356 | -1.160 | 0.260 | 1.038 | 36.9 |
| $\beta_{ELEV}$ | -0.190 | -0.807 | 0.476 | 1.015 | 33.8 |
| $\beta_{NON-WATER}$ | 0.354 | -1.526 | 2.136 | 1.014 | 69.8 |
| $\beta_{NON-WATER^2}$ | -0.051 | -0.745 | 0.616 | 1.033 | 35.7 |
| $\beta_{URBAN}$ | -0.613 | -1.025 | -0.256 | 1.019 | 22.9 |
| $\sigma^2$ | 5.961 | 1.809 | 12.356 | 1.026 | 18.1 |
| $\phi$ | 31.658 | 4.257 | 59.072 | 1.000 | 8.7 |
| $\sigma_T^2$ | 1.878 | 0.505 | 5.083 | 1.057 | 49.4 |
| $\rho$ | 0.006 | -0.387 | 0.376 | 1.001 | 45.0 |
| $\alpha_0$ | -0.614 | -0.935 | -0.296 | 1.042 | 18.0 |
| $\alpha_{NON-WATER}$ | 0.008 | -0.139 | 0.161 | 1.002 | 10.0 |
| $\alpha_{LAT}$ | -0.201 | -0.369 | -0.028 | 1.036 | 10.8 |
| $\alpha_{OBS}$ | 0.026 | 0.000 | 0.052 | 1.030 | 2.3 |
| $\alpha_{MON}$ | 0.204 | 0.036 | 0.373 | 1.009 | 12.0 |
| $\alpha_{MON^2}$ | -1.152 | -1.386 | -0.935 | 1.011 | 11.5 |

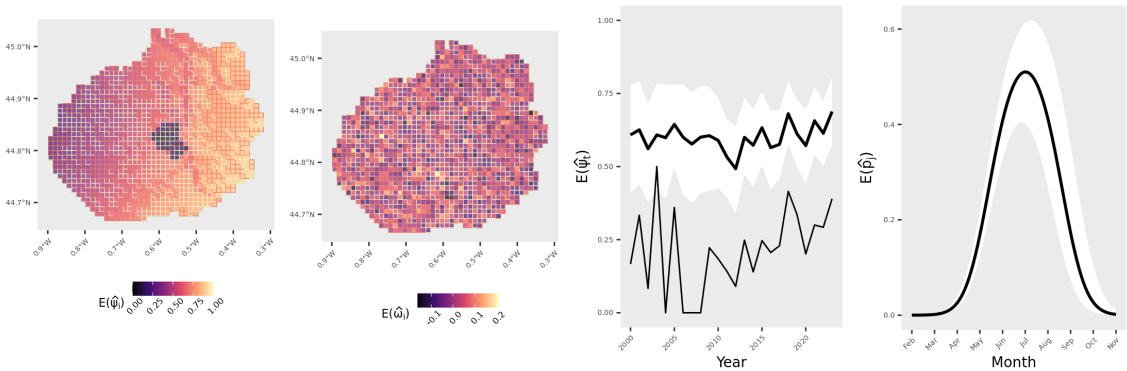


Fig. F.4: Distribution of the meadow brown *Maniola jurtina* in Southwest France (buffer of 10 km$^2$ around Bordeaux). Results for the model with NGG=15 and weakly informative prior for $\phi$. Values plotted in the occupancy map (left) were calculated as $\mathbb{E}(\hat{\psi}_i) = \frac{1}{T}\sum_{t=1}^{T} \hat{\psi}_{it}$, and refer to in-sample estimates for sites with data (cells with yellow borders) and out-of-sample predictions for missing cells (those with red borders). The spatial random effect $\mathbb{E}(\hat{\omega}_i)$ was also derived from in-sample estimates and out-of-sample predictions from the model. Trend plots show the proportion of occupied sites over the 24 years of data, calculated as $\mathbb{E}(\hat{\psi}_t) = \frac{1}{I}\sum_{i=1}^{I} \hat{\psi}_{it}$. The naive occupancy trend (number of sites with detection divided by the total number of sampled sites per year) is depicted by the thinner line. Detection probability over months (secondary periods) was also derived from in-sample predictions. The black line depicts the average detection trend, and the white bands depict the upper and lower bounds of the 95% credible interval. No $1 \times 1$ km cell was 100% covered by water (the maximum was 93%). Parameter values derived from 1,500 posterior distribution draws.

Table F.4: Estimated parameters of the multi-season site-occupancy model with spatial and temporal random effects fitted to data of the meadow brown *Maniola jurtina* (Linnaeus, 1758). Results for models with 15 spatial neighbors, and weak or informative $\phi$ priors. PPO: Prior-Posterior Overlap. PPO: Prior-Posterior Overlap. Abbreviation of covariates = LAT: Latitude, LON: Longitude, ELEV= Elevation, LAND= non-water land cover, GRASS: Cover of natural grasslands, OBS: Number of observers, MON: survey month. The mean, 95% Credible Intervals, and Rhat were calculated using the 1,500 posterior distribution draws of each parameter.

| Parameter | Mean | LCI | UCI | RHat | Overlap |
|---|---|---|---|---|---|
| $\beta_0$ | 0.632 | -0.192 | 1.538 | 1.058 | 44.7 |
| $\beta_{LAT}$ | -0.178 | -0.905 | 0.475 | 1.072 | 36.8 |
| $\beta_{LAT^2}$ | 0.188 | -0.237 | 0.671 | 1.032 | 28.1 |
| $\beta_{LON}$ | 1.305 | 0.744 | 2.041 | 1.025 | 30.2 |
| $\beta_{ELEV}$ | 0.097 | -0.511 | 0.705 | 1.037 | 34.1 |
| $\beta_{NON-WATER}$ | 0.057 | -1.467 | 1.454 | 1.042 | 65.0 |
| $\beta_{NON-WATER^2}$ | -0.061 | -0.629 | 0.461 | 1.048 | 31.3 |
| $\beta_{URBAN}$ | -0.649 | -1.030 | -0.354 | 1.056 | 18.5 |
| $\sigma^2$ | 3.338 | 1.186 | 7.623 | 1.074 | 32.6 |
| $\phi$ | 31.594 | 4.697 | 58.407 | 1.004 | 8.5 |
| $\sigma_T^2$ | 0.468 | 0.158 | 1.261 | 1.001 | 70.0 |
| $\rho$ | 0.031 | -0.363 | 0.412 | 1.019 | 45.1 |
| $\alpha_0$ | -0.065 | -0.444 | 0.288 | 1.012 | 22.0 |
| $\alpha_{NON-WATER}$ | -0.094 | -0.235 | 0.042 | 1.020 | 10.0 |
| $\alpha_{LAT}$ | 0.100 | -0.081 | 0.288 | 1.015 | 12.1 |
| $\alpha_{OBS}$ | 0.045 | 0.017 | 0.077 | 1.001 | 2.5 |
| $\alpha_{MON}$ | 1.188 | 0.919 | 1.488 | 1.002 | 14.6 |
| $\alpha_{MON^2}$ | -3.344 | -3.822 | -2.863 | 1.008 | 5.8 |